Weekly Schedule

Spring 2027 · Mondays are lecture, Wednesdays are tutorials

Mondays are lecture; **Wednesdays** are hands-on tutorials that feed into that week's assignment. Data Safari exercises appear in Weeks 4, 7, 10, and 13. Slides, tutorials, and downloadable assignment templates are on the course website.

Week	Topic	Assignment	Data Safari
1	Why experimental design matters; setting up R, RStudio, and Quarto	PS1	
2	R fundamentals and data wrangling	PS2	
3	Exploring data I: distributions and summary statistics	PS3	
4	Exploring data II: relationships and group comparisons	PS4	Safari 1
5	Principles of experimental design: replication, randomization, blocking	PS5	
6	Design in practice: sketching designs from the published literature	PS6	
7	Hypothesis testing I: the t-test and one-way ANOVA	PS7	Safari 2
8	**Spring Break**		
9	Everything is a linear model: t-tests, ANOVA, and correlation as `lm()`	PS9	
10	Multiple regression, ANCOVA, and interactions; marginal effects	PS10	Safari 3
11	Generalized linear models: binary and count data; chi-square as a linear model	PS11	
12	Non-independence: mixed-effects models I (random intercepts)	PS12	
13	Mixed-effects models II: random slopes, nested vs. crossed designs	PS13	Safari 4
14	When assumptions fail (ranks, robust SEs, permutation) and choosing a model	PS14	
15	**Final Project Presentations**		

From Week 9 onward the course is organized around one idea: **common statistical tests are special cases of the linear model**, and we read every model with the `marginaleffects` package (predictions, comparisons, and slopes in the units of the data).

Graduate project milestones (BIOL 5860 only)

Four checkpoint deliverables spread across the semester, each documenting one stage of your independent project. See the Assignments overview for the full description. As with weekly assignments, click the name to download the blank `.qmd` template.

Due by end of	Milestone	Template
Week 7	Experimental design and planned hypothesis testing	Part 1
Week 10	Data carpentry and exploration	Part 2
Week 13	Statistical modeling	Part 3
Week 14	Visualization	Part 4
Week 15	Final presentation covering the entire process	—

> These due dates are a suggested pacing (each milestone lands a week after the most relevant course content), not fixed by any particular constraint — adjust freely in this table and in the corresponding files' front matter if you'd prefer a different spread.

Notes

- Slides open as a separate reveal.js presentation (use arrow keys or click to advance); tutorials and assignments render as regular pages on this site.
- The graduate-only Project is presented in Week 15, alongside undergraduate project presentations.